Predictive Modeling of London Fire Brigade Response Times and Pump Deployment Patterns: A Ward-Level Machine Learning Analysis.

Bradley Marimbire, Diana Muthoni, Jacob Palakunnathu

Abstract—Accurately predicting response times and the number of pumps deployed in the coming minutes is crucial for fire brigades. This enhancement improves their intervention response time, directly affecting the adequateness of victim attendance. Fire departments in London are facing a rising number of emergency calls and interventions, leading to an increased demand for resources and accurate response times for fire engine arrivals. To minimise damage and enhance safety, fire brigades must consider crucial factors, including the location of their stations, traffic congestion, and the types of properties involved. By effectively addressing these elements, they can ensure timely responses that save lives and protect property. Therefore, this paper aims to implement two separate pipelines to predict Fire Brigade response time and the number of fire brigades dispatched at an incident's call, utilising statistical techniques, unsupervised spatiotemporal, geospatial analytics and supervised machine learning algorithms for prediction. The London Fire Brigade department provided a dataset with temporal and spatial information. In addition, this includes, incident group, property types, number of pumps deployed, the station where the pump was deployed, and response times. The information recorded ranged from January to April 2017. Great efforts were concentrated on data comprehension, cleaning, exploratory data analysis and feature engineering of additional features to aid model understanding. As revealed in the outcome, such methods are mature enough to provide a good solution with a satisfactory margin of error for real-life implementations.

Index words – LFRS, Fire Brigade, Machine Learning, Spatio-temporal, Predictive Analysis

I. INTRODUCTION

Emergency response management presents a significant challenge for fire and emergency services worldwide, including those in London. Through observing recent years, scientists and researchers have employed various statistical, analytical, and artificial intelligence techniques, particularly in the realm of machine learning, to develop sufficient emergency response management (ERM) systems.

Emergency incidents in the various boroughs of London often arise from factors such as the unsafe storage of flammable materials, insufficient escape routes, and problems associated with e-bikes and e-scooters. Additionally, arson and intentional fires, frequently involving waste, contribute to many incidents. The London Fire Brigade is also assigned to address emergencies in relation to floods and other critical situations, including chemical, biological, radiological, and nuclear threats. First responders in the city must be prepared to handle a wide range of emergencies, encompassing fire incidents, rescue operations, and alarm responses.

The importance of this can not be undervalued, as fire and other emergency incidents have massive socioeconomic and sociological consequences, including considerable economic and social upsets to individuals, households, and societies.

Machine learning models that leverage data from a wide range of sources, including public safety records, property inspections and violations, building assessments, tax information, and community surveys, can aid in anticipating the fire risk of commercial buildings across different boroughs within London.

This research aims to explore how the impact of the operational effectiveness of industrial fire brigades can be enhanced by optimising and introducing new key configurable parameters that influence emergency response time and the number of pumps deployed. Since these two factors are affected by various elements, this machine learning study will identify, analyse, and evaluate the extent to which each element impacts the overall performance of the number of pumps deployed and the time the fire brigade will arrive at an incident. The main research objectives are achieved by answering these three questions.

1. What are the most crucial industrial fire brigade parameters required to reduce emergency response time for London fire brigade departments, and what is the direct effect on response time improvement?

2. What are the most crucial factors determining the number of fire brigades deployed for an emergency?

3. Can additional features be modelled to improve performance in predicting fire brigades' number of pumps and deployed and emergency response time?

4. Does the implementation of ensemble modelling through stacking improve the accuracy of fire brigade response time predictions compared to individual machine learning models?

II. LITERATURE REVIEW

Summary of Related Work on Fire Emergency Event Prediction

Several recent studies have focused on modelling fire service response and operation times using statistical and machine learning methods, identifying critical predictors for effective emergency response management. Sharma et al. (2024) utilized negative binomial regression alongside descriptive analytics to analyze fire incidents in Edmonton, Canada. They developed a multi-scale data preparation pipeline and identified key spatiotemporal, socioeconomic, and demographic predictors. Their approach classified neighborhoods by risk and emphasized the impact of external factors, such as COVID-19, on emergency preparedness and response prediction.

Hassler et al. (2024), working with national Fire and Rescue Services data in Sweden, employed GIS-based network analysis and spatial regression to model response times. Their work underscored the importance of incorporating spatial planning and event-specific factors to achieve accurate predictions and enhance service efficiency.

More recently, Sahebi et al. (2025) studied urban firefighting operation times in Ilam City, Iran, through several machine learning models, including GLM, SVM, and Random Forest. They found the Generalized Linear Model (GLM) performed best (RMSE: 14.21 ± 1.65 minutes), with key predictors including personnel numbers, rescue equipment availability, fire engine deployment, and manual extinguishers. Their results stressed the necessity of comprehensive data collection to refine predictive accuracy and optimize urban fire management strategies.

These studies collectively highlight the value of integrating spatial, temporal, demographic, and resource-related features within predictive models to enhance firefighting response effectiveness and preparedness.

III. Methodology

Data Collection: The London Fire and Rescue Service (LFRS) is the busiest in England and one of the largest firefighting and rescue organizations worldwide. For this study, LFRS supplied dataset was used, containing approximately 32,000 intervention records spanning January 1 to April 30, 2017. Each record comprises 32 attributes capturing temporal details (call time, date of call, pump arrival time), incident classification, geographic information (address and home station, eastings and northings), and operational metrics. This comprehensive dataset supports analyses of response efficiency and service demand.

Data Cleaning and Pre-processing: We started by loading the raw intervention records from the provided CSV and performing a thorough audit - checking each column's data type, tallying missing values, and validating the overall schema. From there, we filtered out any non-London incidents and kept only those records with complete pump-count data, removing less than 1 % of the total. To address spatial gaps, we back-filled missing easting and northing values using their rounded counterparts. We then unified all timestamp fragments into a single call_datetime field (formatted to ISO 8601) and stripped away non-essential columns - such as full postcodes, borough and ward codes, and separate date/time fields - to streamline our dataset. This initial cleanup produced a reliable, lean core dataset; from there, we applied further, goal-driven transformations and feature engineering when building our pump-count and response-time prediction models.

We decomposed each call timestamp into hour of day and day of week to capture weekly demand rhythms. For geospatial analysis, we overlaid a uniform 2 km × 2 km grid by integer-dividing each incident's easting and northing coordinates, tagging each cell with a unique grid ID. We then measured the straight-line (Euclidean) distance from central London to each incident and sorted these distances into ten deciles. Plotting average response time against these deciles revealed a clear gradient: median response times remain around 280–300 s in the inner three deciles before rising steadily to over 370 s in the outermost decile, confirming that response efficiency diminishes with increasing distance from the city center, as shown in Figure 1.

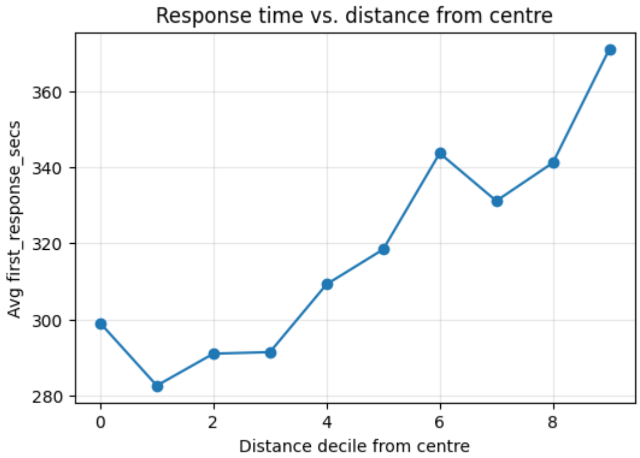

Figure 1: *Response time vs Distance decile*

The weekday–hour heatmap in Figure 2 uncovers pronounced call-volume rhythms. Incidents surge during late mornings (10:00–14:00) and again in the early evenings (17:00–19:00). Saturdays exhibit the highest peak, with volumes exceeding 300 calls in the busiest hour, while weekends sustain elevated activity into late evening. Weekday patterns are similar but less extreme, with midweek (Tuesday–Thursday) mornings slightly quieter than Mondays and Fridays. Overnight hours (00:00–06:00) consistently register the fewest calls, dipping below 80 incidents per hour. These temporal insights guided the inclusion of hour-of-day and day-of-week features in our predictive models.

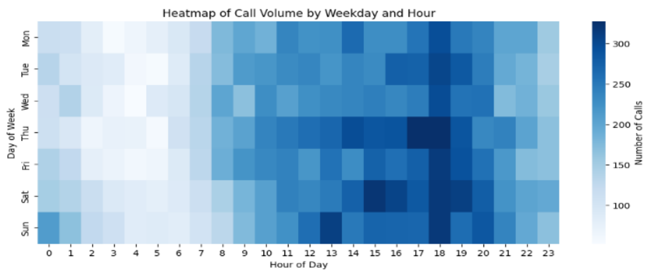

Figure 2*: Call volume Heatmap on Day of week x Hour*

The strong right-hand skew in our first-response times distribution shows most calls falling between 200–400 seconds but with a meaningful tail extending beyond 600 seconds, which prompted us to avoid parametric methods that assume normality. Instead, we applied nonparametric tests better suited to this distribution. Likewise, the pump-count distribution revealed that single- and two-pump dispatches account for roughly 90% of incidents, with three-pump calls already scarce and four-plus pumps exceedingly rare. To maintain statistical reliability and prevent sparse-data issues, we merged all cases involving four or more pumps into a single “4+ pumps” category down in pipeline 1.

We constructed a three-dimensional spatiotemporal “cube” by aggregating, for each borough and each hour of the week (0–167), the average response time, call volume, and median pump count. This structure captures the ebb and flow of demand and performance over both space and time. From this cube, we extracted the 168-dimensional vector of mean response times for each borough as our feature matrix, since it yielded the highest silhouette scores compared to other combinations of metrics.
We evaluated silhouette scores for k = 2 through 10 and found that k = 4 offered the best trade-off between cluster cohesion and interpretability. With four clusters defined, we then visualized their spatial footprint on a choropleth map (Figure 3).
These spatiotemporal clusters illuminate how service performance varies not just by location, but also by hour and day, insights that can directly inform dynamic resource allocation and staffing strategies.

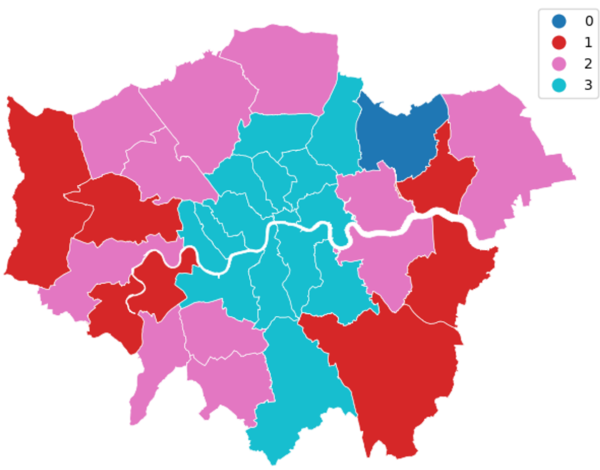

***Figure 3**: Choropleth of London after K-means clustering*

We then applied a nonparametric statistical testing framework directly to the four spatiotemporal clusters to confirm that they represent truly distinct operating conditions. For the continuous outcome (first-response time) and the discrete outcome (number of pumps), we first ran the Kruskal-Wallis H test, which yielded highly significant omnibus results in both cases ($p < 0.001$). We followed each with Dunn’s post hoc pairwise comparisons—using a Bonferroni adjustment to preserve a family-wise α of 0.05—which showed, for instance, that Cluster 1 (outer boroughs) experienced significantly slower median response times than other clusters, and that Cluster 3 (entertainment districts) required more pumps on average than Cluster 2 (inner-suburban areas).

We selected the Kruskal-Wallis test with Dunn’s post hoc comparisons because our initial plots made it clear that both response times and pump counts were heavily skewed, with long tails and outliers, and that group variances varied substantially. These characteristics violate the normality and homoscedasticity assumptions of a one-way ANOVA. By ranking observations rather than relying on raw values, Kruskal-Wallis accommodates non-normal distributions and unequal variances, and Dunn’s test, with Bonferroni adjustment, then delivers pairwise comparisons without the need for further assumptions.
The same nonparametric testing framework was adopted to study the distribution of key categorical features. The relationships between them further provided clarity on how to select and engineer features further down the following pipelines.

Pipeline 1: Prediction of the number of pumps deployed.

This section outlines the methodology used to develop a predictive model for the number of fire pumps deployed by the London Fire Brigade. The main goal was to build an accurate and generalisable machine learning model to predict the number of fire pumps dispatched at the time of an emergency call, using spatial, temporal, and property-related attributes such as borough location, coordinates, incident group, property type, and time of call.

Feature selection: To ensure accurate and realistic prediction, features likely to introduce target leakage were excluded from the modelling process. These include variables that are either a consequence of the target or only become available after the pumps have been dispatched. For example, the number of stations with pumps attending is directly derived from the target variable number of pumps, making its inclusion redundant and potentially misleading. Similarly, the first pump arriving at attendance time and first pump deployed from station reflect outcomes recorded after pumps have already been dispatched and therefore would not be known at the time of decision-making. The special service type is only confirmed after an incident assessment, and the response time quantile derived from actual arrival times, which depend on how many pumps were sent. Additionally, incident station ground may implicitly indicate the scale or type of response and could act as a proxy for the target, while grid ID may capture highly localised deployment behaviours that lead to information leakage. Removing these variables ensures the model relies only on information available at the point of the emergency call, supporting valid generalisation. Including these features would result in target leakage, which can lead to unrealistically high model performance and poor generalisability in real-world applications.

Feature engineering: Based on the exploratory analysis of the pump count distribution, the continuous pumps variable was converted into a categorical feature. This grouped responses into 1, 2, 3, and 4+ pumps. The grouping of 4+ into a single category was justified due to its sparsity and to improve model stability in underrepresented classes.

Encoding: To make the dataset suitable for modelling, label encoding was applied to the target column (number of pumps) for model compatibility, especially XGboost to convert the string categorical variables to numeric. One hot-encoding was used for input categorical features to avoid ordinal misinterpretation and to allow models to treat each category independently.

Train-Test Split and Class Imbalance Handling: The train-test split dataset was stratified to preserve class distribution. The dataset was split into training and testing sets using stratified sampling, which ensures the class distribution of the number of pumps is preserved. This is critical for addressing class imbalance, particularly because the 4+ class forms a small minority. Though SMOTENC was initially considered for oversampling, it was discarded due to its negative impact on XGBoost's performance, as synthetic instances disrupted the model's learning on minority classes.

Model Selection and Hyperparameter Tuning: Three models were trained: Logistic Regression, Random Forest Classifier and XGboost. Each model was tuned using RandomisedSearchCV with StratifiedKFold (5 splits) cross-validation to optimise hyperparameters and prevent overfitting. The RandomisedSearchCV was selected for its efficiency in exploring a wide hyperparameter space without the computational cost of an exhaustive grid search. The tuning was guided by the accuracy score as the primary selection metric. The metric selected the best performing hyperparameter combination during cross-validation. StratifiedKfold preserves the class distribution of the multiclass target.

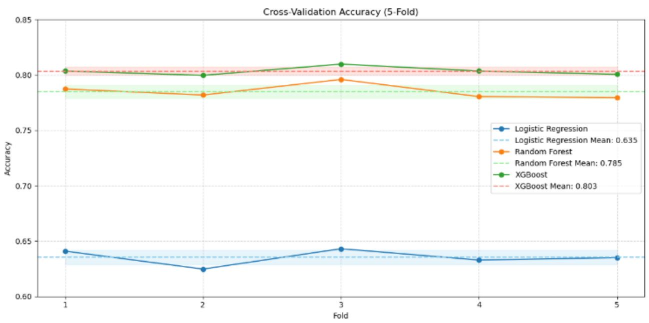

Figure 4*: Cross-validation performance analysis on the models*

To ensure that the models yielded the same results a randomness control of randomstate=42 (seed) was applied consistently for replicable results when rerun. The pipeline also incorporated ColumnTransformer for numeric and categorical preprocessing in a unified structure.

Evaluation metrics: Models were evaluated using accuracy, precision, recall, F1-score, confusion matrices and Per-class recall plots. The confusion matrices (Figure 5) demonstrates the comparative performance of each model. XGBoost exhibited superior predictive ability, especially in handling the minority 3 and 4+ classes, while Logistic Regression showed the highest misclassification rate.

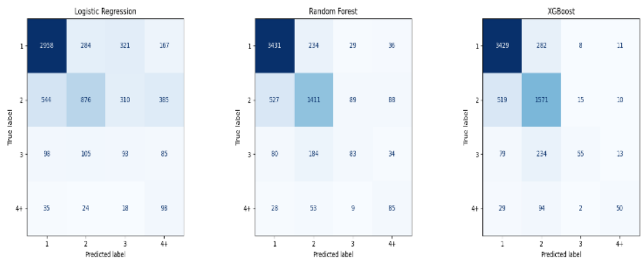

Figure 5*: Confusion matrices for the models*

Pipeline 2: Prediction Of Response Time

Missing Value Treatment: For the first_response_station column, rows with missing values were replaced with "Unknown_station" rather than being dropped. We implemented a borough-based median imputation for the unknown station's associated with missing response times (first_pump_arriving_attendance_time).Fire-brigade response times are primarily influenced by geographic location; incidents in the same borough typically experience similar response times, regardless of which specific station responds. We specifically chose to use the borough-level median response time rather than the mean because response time data often contains outliers. The median provides a preferred measure of central tendency that is not skewed by these extreme values. This exact procedure was implemented for our dataset's average response time per bin column, but instead of median, mean was used. This approach acknowledges the London Fire Brigade's real-world operational patterns, where station coverage areas within the same borough tend to have similar baseline response characteristics.

Outlier Treatment: The outlier treatment approach involved addressing questioningly low values in our target variable. The distribution of response times consisted of values that required careful consideration and management to preserve real-world operational patterns. Based on domain knowledge from the London Fire Brigade documentation [3], we established that response times below 60 seconds were operationally unattainable. Therefore, we implemented lower-bound capping (Winsorizing), which in turn dismissed values that didn't make logical sense. For the upper range, we observed maximum response times of approximately 1196 seconds (19.9 minutes). After analysing the operational context, we determined these values were legitimate because incidents could have been in remote locations, there could have been a response during high-traffic periods, or there were areas with challenging access conditions. We deliberately preserved these data points to maintain the real-world distribution pattern. This decision was crucial for developing a model to accurately predict longer response times, often representing the most urgent cases requiring functional improvement.

Feature engineering: In our experiment feature engineering was used to attempt to improve the predictive power of our model. Our procedure incorporated historical context and operational knowledge from the London Fire Brigade operations. Analysis of the "Fire Facts - Incident Response Times 2017" report [3] revealed that prior to the Fire and Rescue Services Act 2004, London's emergency response infrastructure was configured based on four risk categories (A-D) determined by property characteristics and area density. Although these categorical standards were officially replaced with an "equal entitlement" principle, we hypothesised that the physical infrastructure established under the previous system would resume to influence current response patterns. To capture these historical influences, we engineered two new categorical features. '*Area Density Class'*, this feature categorised locations into four density types that mirror the historical risk categorisation system. *'Property Usage'* was the second classification, this feature consisted of classified properties into functional categories reflecting the different priority levels historically assigned to various property types. Following this, we reclassified all "False Alarm" instances to simply "Alarm" in the incident_group variable. This decision was based on the validity that at the time of dispatch, fire brigades only knew they were responding to an alarm, and they could not know in advance whether it would turn out to be a false alarm. This recording ensured that our predictive models reflected the information available at the moment of dispatch decision-making. For temporal variables, we implemented cyclical encoding to preserve the cyclic nature of time-based features. This comprehensive approach to temporal feature engineering ensured that all cyclical time patterns were appropriately represented in our model, avoiding artificial discontinuities. Spatial features were also engineered when observing the emergency response time; the physical distance between fire stations and incident locations is critical to analysing emergency response times. We developed a process to construct spatial features that greatly improved our model's predictive capabilities while handling missing data.

Following this, we identified the easting and northing coordinates for all London Fire Brigade stations using the GridReferenceFinder tool [4]. The procedure for this involved converting fire brigade station postcodes into precise geographic easting and northing coordinates. From this, additional distance-based features were engineered utilising the Euclidean distance formula. To prepare categorical variables for our predictive models, we applied appropriate encoding techniques to transform them into a numerical format. The first approach we used for encoding our variables was One-hot encoding; this was done for the following features, 'property_category', 'area_density_class', 'property_usage', and 'incident_group'. These features had distinct values ranging from 1 to 10; this approach was proper to convert categorical variables into a format suitable for machine learning algorithms that cannot directly process categorical data. Boolean columns were also transformed into binary integers to ensure compatibility with our machine-learning algorithms. Many categorical variables in our dataset consisted of high cardinality, which would create excessive dimensionality if one-hot encoding was used. To address this challenge, we implemented target encoding for these features.

Feature selection and removal: We removed variables that could introduce bias, redundancy, or unnecessary complexity to our model. We removed the postcode_district column because it infers the same information as borough_name. The address_qualifier column consisted of values filled after dispatch was reached. Since we could not guarantee that this information was available at dispatch time, keeping it would introduce a form of data leakage; thus, the column was removed. The special_service_type column contained 69% missing values; imputing missing values for a categorical variable like this would likely introduce significant bias. As a result, this column was also removed.

Dimensionality reduction: To address the high cardinality of the property_type column with 258 distinct values, we opted to remove it and retain the more manageable property_category column. Similarly, we replaced the stop_code_description column with the incident_group column, as it offers a higher-level abstraction of incident types, reducing dimensionality while preserving crucial categorisation information. This approach balanced information preservation with model complexity, supporting predictive performance and interpretability.

Data Preparation for Modelling: Before training the model, we applied a logarithmic transformation to the target variable, first_pump_arriving_attendance_time, which exhibited a right-skewed distribution. To address this skewness and improve model performance, we used the log1p function, which handles zero values effectively. This transformation helped stabilise variance across the data. When evaluating model performance and generating predictions, we converted the log-transformed values back to their original scale, ensuring that predictions were expressed in meaningful time units (seconds) relevant to fire brigade operations.

Model Selection and Implementation: Given the nature of our dataset, we decided to use tree-based algorithms rather than traditional linear regression models. Our dataset contained a large number of binary encoded columns, One-hot encoded features representing categorical variables and numerical mappings like Easting and Northing coordinates. Tree-based algorithms were better suited for this kind of data because they can handle this type of data without requiring feature scaling or transformations. They can also capture non-linear relationships and feature interactions automatically, which would be difficult for a linear regression model to learn without heavy manual feature engineering. Lastly, they are more preferred when dealing with irrelevant features, as trees can ignore unimportant splits during training. We implemented and compared two tree-based algorithms for feature importance. This approach provided considerable benefits, as it confirmed the value of our engineered features, particularly the distance-based and temporal features. Lastly, it provided insight into the operational factors that had the greatest impact on fire brigade response efficiency. The consistency between different tree-based models in identifying important features reinforced our confidence in the selected predictors and the overall modelling approach.

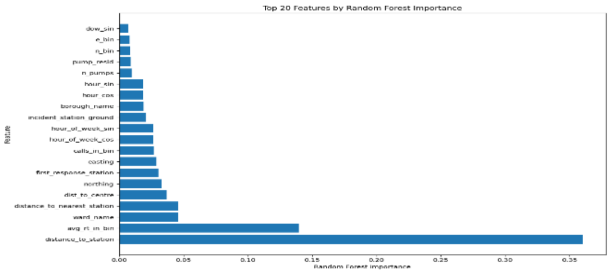

Figure 6*: Top 20 Features by Random Forest Importance*

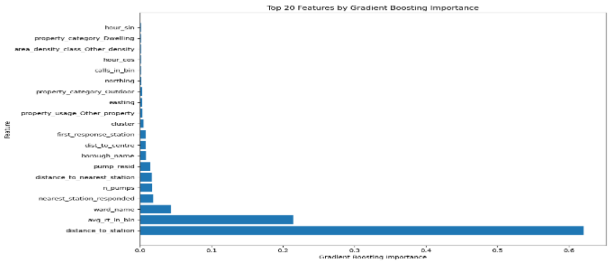

Figure 7*: Top 20 Features by Gradient Boosting Importance*

Model Evaluation and Selection: After identifying key features, we evaluated tree-based regression models to find the best approach for predicting fire brigade response times. Four machine learning algorithms, Random Forest, Gradient Boosting, XGBoost, and LightGBM, were evaluated to determine the most suitable predictive model. These models were selected for their strong performance in regression tasks and ability to handle complex, non-linear relationships. All models were trained using a subset of features selected based on Random Forest feature importance rankings. Model performance was assessed using both test-set metrics and 5-fold cross-validation. The following metrics used were mean Absolute Error (MAE) to capture average prediction errors, Root Mean Squared Error (RMSE) to penalise larger errors more heavily, R^2 Score to quantify variance explained by the model and Cross-validated RMSE (CV RMSE) to evaluate generalisation performance.

*Ensemble Modelling with Stacking***:** To further improve prediction accuracy, we implemented an ensemble approach using stacking, which combined the strengths of multiple base models. The stacking approach worked by training multiple base models: Random Forest, Gradient Boosting, XGBoost, and LightGBM, using cross-validation to generate predictions from each base model. In addition to this a meta-learner (Ridge regression) was used to train on the base model predictions to produce the final output. We evaluated the stacking ensemble using the same metrics and cross-validation approach as the individual models. The stacking ensemble approach was crucial, as it mitigates the weaknesses of individual models by combining their strengths; it reduces overfitting risk through the cross-validation mechanism in the stacking process, and it captures different aspects of the data that single models might miss. This approach provides more stable predictions across different types of incidents. After implementing the stacking ensemble, we compared its performance against all individual models to determine whether it offered improvement over the best single model.

IV. Results

Pipeline 1: Prediction of the number of pumps deployed.

Table 1: Comparison of Model Performance for Predicting Fire Pump Deployment

Model	Overall Accuracy	Macro Precision	Macro Recall	Macro F1-Score
XGBoost	79.8%	0.712	0.523	0.559
Random Forest Classifier	78.3%	0.584	0.573	0.568
Logistic Regression classifier	62.9%	0.438	0.503	0.425

XGBoost has the highest overall accuracy and macro precision. It performs consistently across the majority and minority classes. Its slight drop in macro recall suggests it occasionally misses some true positives for minority classes (3, 4+), but overall balance is strong.

Random Forest Classifier, accuracy is slightly lower than XGBoost. It performs better in terms of recall compared to XGBoost since it captures more true positives, but at the cost of more false positives (lower precision). This suggests that it generalises well but may misclassify some instances near class boundaries, resulting in slightly reduced precision.

Logistic Regression has the lowest performance across all metrics since it struggles with both majority and minority classes. The model may be underfitting due to its linear nature, which limits its capacity to model complex patterns present in the multiclass data.

Pipeline 2: Predicting first pump arrival attendance time.

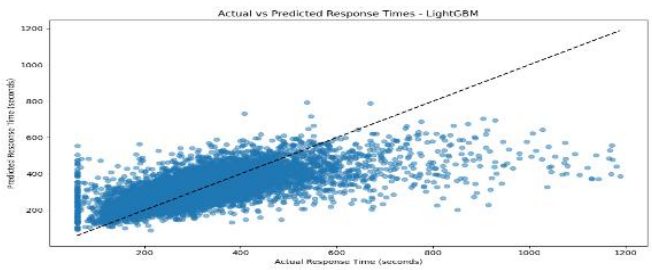

Figure 8*:Actual vs Predicted Response Times - LightGBM*

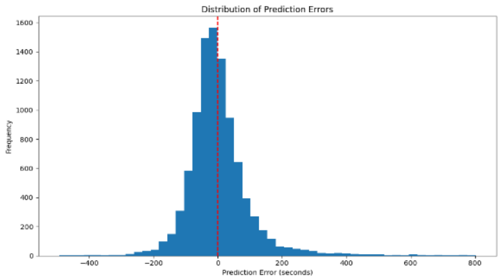

Figure 9*: Distribution of Prediction Errors*

Pipeline 2: Predicting first pump arrival attendance time.

Table 2: Comparison of Model Performance for Prediction of First Pump Arrival Attendance Time

Model	MAE (second)	RMSE (seconds)	CV RMSE (seconds)	R^2
Random Forest	66.58	101.10	98.68	0.4292
Gradient Boosting	63.00	100.04	97.08	0.4411
XGBoost	65.26	100.19	97.53	0.4394
LightGBM	65.53	99.44	96.58	0.4478
Stacking Ensemble	63.88	98.56	95.71	0.4575

Our feature engineering revisions produced significant performance progress across all models, with R^2 values ranging from 0.43 to 0.4575 compared to previous values of 0.34-0.38. The Stacking Ensemble achieves the best overall performance with the lowest RMSE (98.56 seconds) and highest R^2 (0.4575), while the Gradient Boosting model deliveres the lowest MAE (63.00 seconds). The consistently strong cross-validation metrics indicate good generalisation to unseen data.

Figure 8. presents a scatter plot of actual versus predicted response times for the LightGBM model. The visualisation reveals systematic prediction patterns across the response time spectrum. The model tends to overestimate response times for shorter incidents (actual times less than 200 seconds), with predictions clustered above the reference line. In contrast, the model underestimates response times for extended incidents (greater than 600 seconds), with predictions predominantly below the diagonal. The middle range (200-600 seconds) reveals increased prediction variance, indicating that while the model effectively handles typical incidents, it still struggles to fully capture the complex factors driving longer response scenarios. This pattern likely reflects operational realities where extended responses involve unpredictable elements not fully represented in our feature set.

Figure 9. displays the distribution of prediction errors, showing an approximately symmetric bell shape centred near zero, which confirms unbiased predictions overall. Most errors fall within ±150 seconds, with the highest concentration between -75 and +75 seconds, aligning with the improved MAE of 65.53 seconds for the LightGBM model. The error distribution shows slightly better symmetry than previous iterations, with a more pronounced peak around zero, suggesting the enhanced feature set has improved prediction consistency. With approximately 75% of predictions within one minute of actual response times, the model provides practical accuracy for operational planning pursuits. The significant improvement in model performance validates our feature engineering approach and indicates that incorporating additional contextual features can significantly enhance predictive accuracy for emergency response time modelling.

V. Discussion and Conclusions

By cleaning and exploring the data in depth, we've uncovered what really drives London's fire service response times and how resources are deployed. We identified clear rush-hour spikes and geographic hotspots that point to when and where extra support is needed most. Our borough clustering showed that different areas follow unique performance patterns which is proof that a one-size-fits-all strategy wouldn't be enough. The nonparametric tests we ran also highlighted which incident types and pump-count categories truly affect response times, giving us concrete features to feed into our models. This solid analytical foundation paves the way for building machine-learning tools to predict demand and fine-tune resource allocation, ultimately helping the brigade respond faster and smarter.

Pipeline 1: The study for prediction of the London Fire Brigade number of pumps deployed at the time of an emergency was designed using a combination of spatial, temporal and property related features. This was a multiclass classification with class imbalance that leveraged heavily on hyperparameter tuning with stratified cross validation to ensure generalisable model performance. Among the models tested, XGboost yielded the best performance of 79.8% outperforming others in macro precision and macro F1-score. This work demonstrates that tree-based classifiers like extreme gradient boosting can effectively assist in real-time decision making for fire brigade deployment with careful handling of feature engineering, target leakage and model validation.

Pipeline 2: This investigation for pipeline 2 successfully produced predictive models for London Fire Brigade response times by leveraging additional feature engineering techniques and hyperparameter tuning for our machine learning algorithms. We assembled distance-based elements and other notable features, resulting in a significant increase in model performance. By comparing multiple Tree-based regression approaches, we identified the Stacking Ensemble as the optimal model, achieving an MAE of 63.88 seconds, RMSE of 98.56 seconds, and R^2 of 0.4575. The LightGBM model also performed well, with the lowest cross-validation error (CV RMSE: 96.58 seconds) and strong R^2 (0.4478). The improvement in R^2 values from ~0.36 to ~0.4575 validates our feature engineering procedure and demonstrates the model's increased utility for resource planning and performance assessment in fire brigade service operations.

Future Directions pipeline 1: Merging the incident dataset with traffic data, weather conditions, and road network information could address key gaps in the current feature set.

VI. RECOMMENDATIONS AND TEAMWORK

This project explored the prediction of London Fire Brigade response time and the number of pumps dispatched using machine learning models enhanced by extensive feature engineering and spatiotemporal analysis. Key strengths of the pipelines included careful handling of class imbalance, target leakage prevention, thorough preprocessing, feature engineering using documentation, external resources and hyperparameter tuning through various cross-validation techniques. Despite these successes, some limitations were encountered. The dataset did not contain real-time variables such as live traffic conditions, weather data, or road network disruptions, all of which could significantly influence emergency response outcomes. Moreover, while class imbalance was managed through stratified sampling, minority classes (e.g., "4+ pumps") remained underrepresented and challenging to model accurately.

VII. CONTRIBUTIONS

Bradley Marimbire – Introduction, abstract, Pipeline 2 (coding, methodology, results, discussion, recommendation and conclusion).

Jacob Abraham palakunnathu – Literature review, methodology, discussion recommendation and conclusion. Preprocessing (EDA, feature engineering, cleaning, unsupervised machine learning).

Diana Muthoni – Literature review, Pipeline 1 (coding, Methodology, results, discussion, recommendation and conclusion).

VIII. REFERENCES

[1] A. Mukhopadhyay, G. Pettet, S. M. Vazirizade, D. Lu, A. Jaimes, S. E. Said, H. Baroud, Y. Vorobeychik, M. Kochenderfer, and A. Dubey, ‘‘A review of incident prediction, resource allocation, and dispatch models for emergency management,’’ Accident Anal. Prevention, vol. 165, Feb. 2022, Art. no. 106501.

[2] Dilli Prasad Sharma, Nasim Beigi-Mohammadi, Geng, H., Dixon, D., Madro, R., Emmenegger, P., Tobar, C., Li, J. and Leon-Garcia, A. (2024). Statistical and Machine Learning Models for Predicting Fire and Other Emergency Events in the City of Edmonton. IEEE access, pp.1–1. doi:https://doi.org/10.1109/access.2024.3390089.

[3] “Fire Facts – Incident response times 2017” from the London Datastore: Greater London Authority (GLA). (2018), *Fire Facts – Incident response times 2017*, [dataset], London Datastore. Available at: https://data.london.gov.uk/dataset/incident-response-times-fire-facts [Accessed: 1 May 2025].

[4] Grid Reference Finder. (n.d.). *Grid Reference Finder*. Available at: https://gridreferencefinder.com/ [Accessed 2 May 2025].

[5] Hassler, J., Andersson Granberg, T. and Ceccato, V. (2023). Socio-spatial Inequities of Fire and Rescue Services in Sweden: An Analysis of Real and Estimated Response Times. Fire Technology, 60(1), pp.193–212. doi:https://doi.org/10.1007/s10694-023-01496-3.

[6] Sahebi, A., Hojjat Sayyadi, Bahram Havasy and Yousef Veisani (2025). Predicting firefighting operation time in urban areas using machine learning: identifying key determinants for improved emergency response. Discover Applied Sciences, 7(4). doi:https://doi.org/10.1007/s42452-025-06703-0.

[7] H. Kaur, H. S. Pannu, and A. K. Malhi, ‘A Systematic Review on Imbalanced Data Challenges in Machine Learning: Applications and Solutions’, ACM Comput. Surv., vol. 52, no. 4, pp. 1–36, Jul. 2020, doi: 10.1145/3343440.